VIII. Mini Project - Electricity Consumption Analysis

AIM:

To clean, process, and visualize electricity consumption data using Python, Power BI, and Tableau, and to build clear visualizations for identifying consumption trends, regional variations, and consumer usage patterns.

DOMAIN:

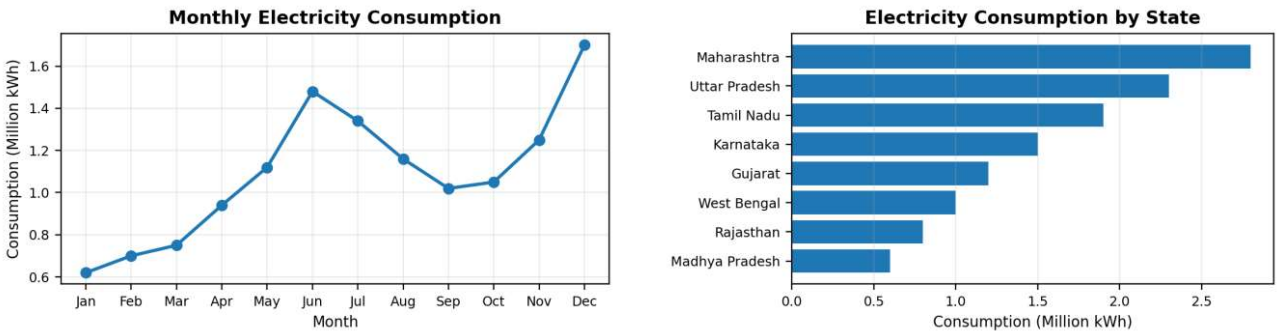
Electricity Consumption Analytics

PROBLEM STATEMENT:

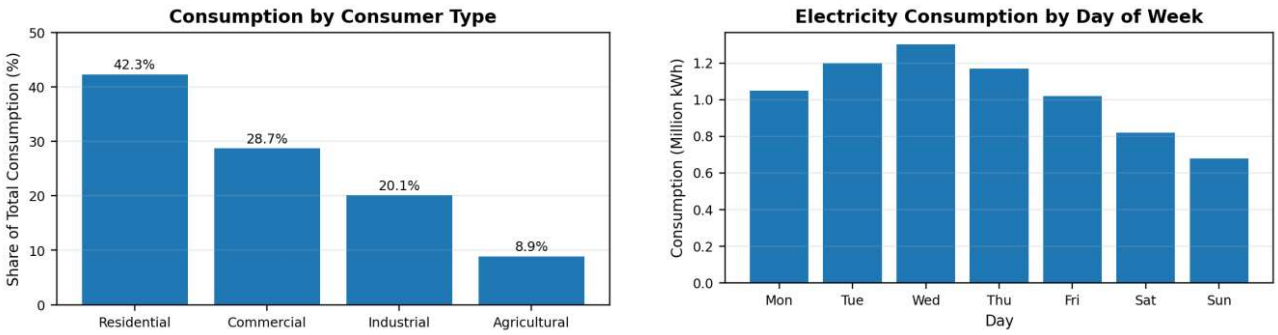
With increasing electricity usage across different regions and consumer categories, it is important to understand how electricity consumption changes over time and across different areas. There is a need for a data-driven visualization system to compare electricity usage by month, state, and consumer type and present the results through interactive dashboards for easier analysis.

OUTPUT:

Python:

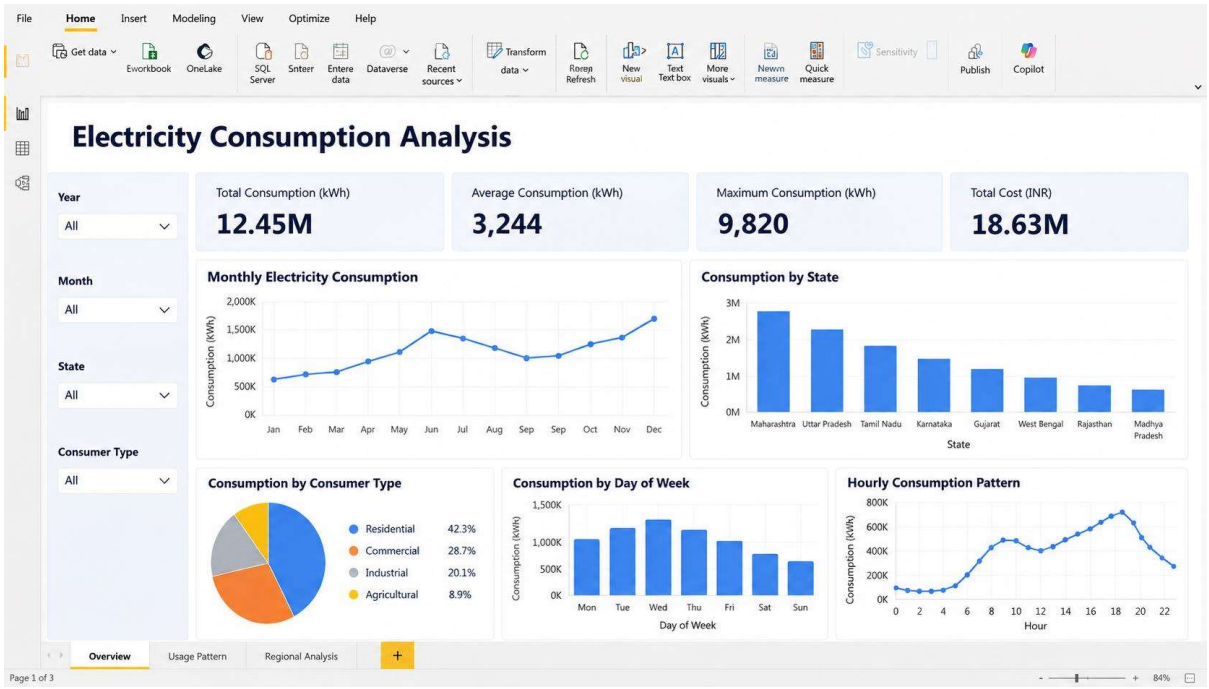


Monthly electricity consumption and state-wise consumption visualizations generated using Python.



Consumer-type and day-of-week electricity consumption visualizations generated using Python.

PowerBI:



Power BI dashboard for Electricity Consumption Analysis.

Tableau:

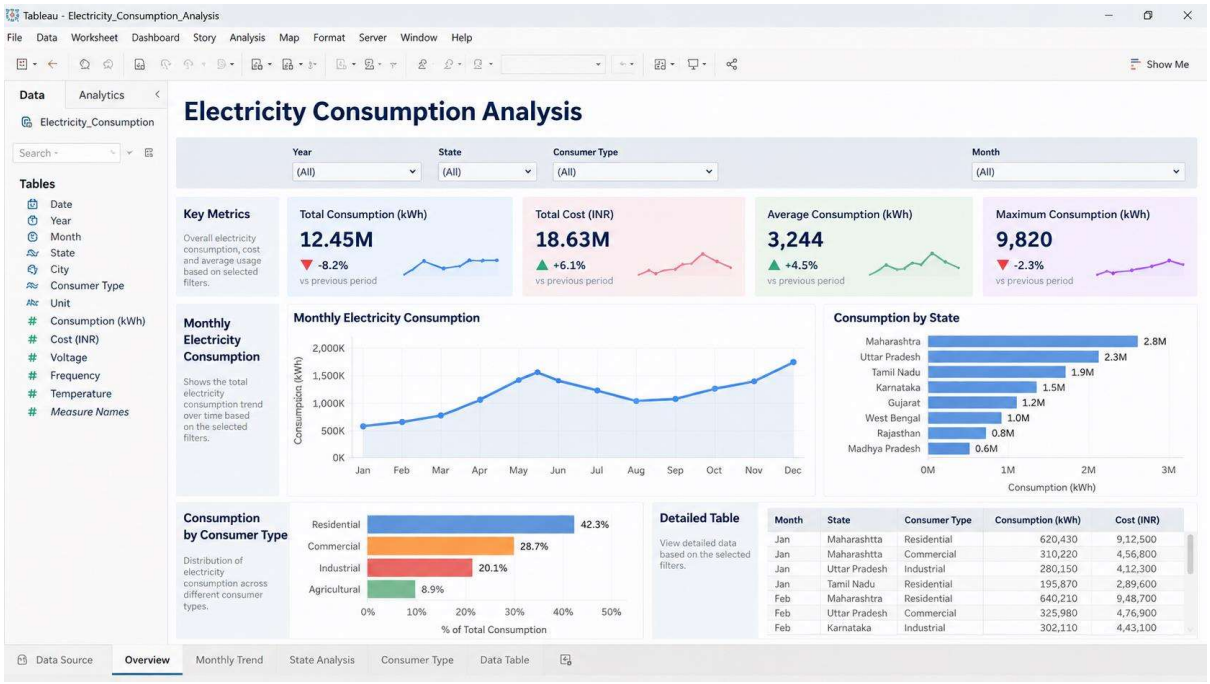


Tableau dashboard for Electricity Consumption Analysis.

RESULT:

Thus, the electricity consumption data was successfully analyzed and visualized using Python, Power BI, and Tableau. The visualizations and dashboards provide a clear view of consumption trends, regional differences, consumer usage patterns, and detailed electricity consumption information.